Muhammad Zaryab Safder

AI and software engineer | BSc Computer Science (2026)

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PROFILE

Computer science graduate who builds AI systems end to end (web and mobile front ends, backend, deployment). Co-built a camera-based fall detection platform as an industrial research and development final-year project; now an AI engineer on LLM-based tools. Seeking a research-oriented master's or research internship in software engineering for AI and security.

Research interests: software engineering for AI, LLMs for code, empirical software engineering, reliability of AI systems, cyber security (software supply chain, malware detection).

EDUCATION

BSc Computer Science, COMSATS University Islamabad, Lahore Campus | 09/2022 to 07/2026

CGPA 3.08/4.00. Final-year project graded A. Coursework: Machine Learning, Artificial Intelligence, Computer Vision, Parallel and Distributed Computing, Operating Systems, Algorithms, Databases, Linear Algebra, Statistics.

PROJECTS

Eldercare: patient monitoring and fall detection platform | final-year project, industrial research and development, two-person team | supervisor Ms. Asmara Safder

- **My part, the system around the model:** Flutter web and Android app, FastAPI backend, self-hosted PostgreSQL and Redis, alerts through WebSockets and Firebase push notifications, Docker deployment, integration testing, and most of the project report.
- **Team result:** the detection model, MAGnet (a multimodal attention graph network, built and trained by my teammate Muhammad Hasaan), reached 98.2% accuracy and 99.78% ROC AUC on 38,674 windows from two public datasets (NTU RGB+D 120, FallVision). Falls only; seizure detection is planned future work.
- Co-author of a research manuscript based on the project (in final stages of writing).

Empirical software engineering projects | github.com/Zaryab111

- **bugfix-tests-mining**: mined 12,537 commits from 15 Python projects to ask whether bug fixes come with tests. The fix-keyword rule was 30% precise on a labelled sample; the first result (50.0% vs 46.2%) did not survive a fair comparison (41.7% vs 38.8%, p = 0.059).
- **rest-api-lint**: checker for REST API design rules, run on 200 public OpenAPI specifications. Precision was 39% on 100 flagged issues checked against a written rule; after fixing the noisy rules it was 89% on a fresh 100.
- **python-logging-study:** static analysis (ast) of logging practice in 21 Python projects; applications log about 8 times more than libraries (2.24 vs 0.28 calls per 1000 lines, p = 0.002).
- **pypi-vulnerability-exposure**: the 300 most downloaded PyPI packages against the OSV database; 33.7% have a known advisory and 29.3% depend directly on one, with only a weak link to dependency count (rho = 0.19).

Other projects: parallelised malware detection pipeline; number plate detection with YOLOv5; web and PDF scraping tools.

EXPERIENCE

Associate AI Engineer, Applicon Soft, Lahore | 07/2026 to present

- Set up an LLM-based workflow (Claude) that generates test schemas, database migrations and test suites, and used it to migrate an existing app's backend from Zend to Laravel within a limited token budget.
- Added a speech-enabled chatbot to an existing app (strict instructions, temperature zero) and worked on an agentic retrieval-augmented generation setup with guardrails.
- Built a microphone app that gives pronunciation and accent feedback through a speech API (Mac and Windows).

Freelance content writer and social media manager, remote | 2020 to 07/2026

- Six years of content and social media work: research-based writing for psychiatrists, psychologists and therapists, and the full social media cycle across Instagram, Facebook, LinkedIn, TikTok, YouTube and Pinterest (posts, carousels, reels, captions, scheduling, comment automation, analytics, paid ads).
- Built AI-assisted production pipelines for video and posts: Claude-written scripts, word-level transcripts, scripted After Effects and Premiere Pro workflows and AI-generated b-roll.

SKILLS

- **Languages:** Python, Java, C++, Dart, JavaScript
- **Backend and infrastructure:** FastAPI, PostgreSQL, Redis, WebSockets, Docker, Firebase
- **Mobile and web:** Flutter (one codebase for web and Android)
- **AI engineering:** LLM prompting, retrieval-augmented generation, agentic workflows, LangChain, guardrails
- **Research and data:** mining software repositories, pandas, SciPy, matplotlib, pytest, git and GitHub
- **Video and social media**: After Effects, Premiere Pro, CapCut, Canva, Meta Ads

CERTIFICATIONS AND LANGUAGES

Cisco Networking Academy: CCNA Introduction to Networks (2024). **Coursera:** Machine Learning (Andrew Ng); IBM Mobile App Development. **Meta:** Digital Marketing Associate (2026).
English: IELTS Academic 7.5 (C1). **Urdu:** native.